

Khulna University of Engineering & Technology
B.Sc. Engineering 1st year 1st term (Regular) Examination, 2017
Department of Materials Science and Engineering
MSE 1101
Introduction to Materials Science and Engineering

Time: 3 hours

Full Marks: 210

N.B.: (i) Answer **ANY THREE** questions from each section in separate script.

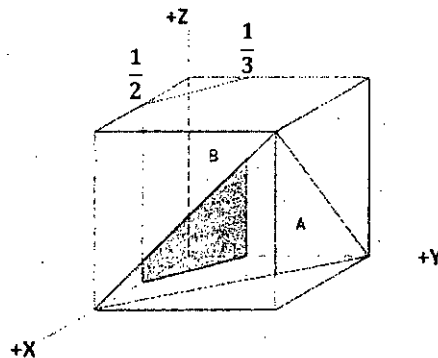
(ii) Figures in the right margin indicate full marks.

Section A

(Answer **ANY THREE** questions from this section in **script A**)

- Q1.** (a) Sketch and describe attractive, repulsive and net force versus interatomic separation for two atoms or ions. Write down the equation of "Equilibrium force" from the plot. (5+7+3)
- (b) What is covalent bond? Explain the formation of covalent bond with example. (3+7)
- (c) Determine the force of attraction between a Ca^{2+} and an O^{2-} ions, the center of which are separated by a distance of 1.25 nm. (10)

- Q2.** (a) Define unit cell. Draw atomic arrangement of a body centered cubic unit cell and show the closed packed direction in this system. (3+5+2)
- (b) What is Miller indices? Determine the miller indices for the planes **A** and **B** shown in the following unit cell: (2+8)



- (c) What is atomic packing factor (APF)? Calculate the APF of a face centered cubic crystal system. (3+12)
- Q3.** (a) What is Fermi energy? Sketch the electron energy band diagram of metal, semiconductor, and insulator and discuss the electrical conductivity based on the energy band diagram. (2+4+6)
- (b) Explain how electron carrier concentration varies with temperature in silicon that has doped with 10^{21} m^{-3} of donor impurity, and in intrinsic silicon. (10+5)
- (c) Calculate the charge carrier concentration of silicon. [Given: Band gap 1.1 eV; electrical conductivity $4 \times 10^{-4} (\Omega\text{-m})^{-1}$; electron mobility $0.14 \text{ m}^2/\text{V.s}$; hole mobility $0.05 \text{ m}^2/\text{V.s}$] (08)
- Q4.** (a) What is ferromagnetism? Discuss the possible sources that contributes in originating ferromagnetism in materials. (3+12)
- (b) What is NDT? List the different techniques of NDT. Explain the basic principle of the ultrasound testing method. (2+3+7)
- (c) A copper wire 15 m long is cooled from 40°C to -9°C , how much change in length will it experience? [The coefficient of thermal expansion is $16.5 \times 10^{-6} (^\circ\text{C})^{-1}$] (8)

Section B

(Answer ANY THREE questions from this section in script B)

- Q5. (a) Explain why a material is deformed at lower applied stress than its theoretical yield stress. (10)
- (b) What are the conditions of substitutional solid solution of impurities atom in crystal? (05)
- (c) What is strain hardening? Explain how mechanical properties (hardness, strength and ductility) of a metal are changed during strain hardening process. (5+10)
- (d) Write the names of crystal defects that are commonly found in crystalline solids. (5)
- Q6. (a) What is modulus of elasticity? Explain the significance of it during selection of materials. (10)
- (b) What is ductility and toughness? Draw the stress-strain curve of the most ductile and toughest material. (10+5)
- (c) Explain, why engineering stress and strain are used in conventional process instead of true stress and strain of a material. (10)
- Q7. (a) What are the common reasons of materials failure? What is your role as a materials engineer in preventing the failures? (5+5)
- (b) Define ductile and brittle fracture of materials. Write the differences between them. (10+5)
- (c) Illustrate the fatigue failure of materials with the help of stress-cycle (S-N) curve. (10)
- Q8. (a) Explain, how color appears in non-metallic materials. (07)
- (b) What is ceramics? Write some common properties of ceramics materials. (3+5)
- (c) A continuous and aligned glass fiber reinforced composite consists of 35 vol. % glass fiber and balance of polyester resin. The modulus of elasticity and tensile strength of the fiber and matrix are given below: (5+5)

	Modulus of Elasticity (GPa)	Tensile strength (MPa)
Glass fiber	69	3500
Polyester resin	3.4	65

Assuming that the stress is applied parallel to the direction of fiber aligned and the matrix stress at composite failure is 10 MPa.

(i) Compute modulus of elasticity of the composite.

(ii) Compute the tensile strength of the composite.

Given, $V_m + V_f = 1$; $\sigma_c = \sigma_m V_m + \sigma_f V_f$; $E_{cl} = E_m V_m + E_f V_f$

$$E_{ct} = \frac{E_m E_f}{E_m V_f + E_f V_m}; \sigma_{cl} = \sigma'_m V_m + \sigma'_f V_f$$

The signs are indicating their usual meaning.

- (d) Briefly explain any one of the following (i) Glass fiber composite (ii) LASER (iii) Optical fiber (10)

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Math 1127

Differential and Integral Calculus

Time: 3 hours

Full Marks: 210

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(ii) Figures in the right margin indicate full marks

Section A

(Answer **ANY THREE** questions from this section in **script A**)

- Q 1. (a) Calculate $\int \frac{dx}{(x+1)\sqrt{1+2x-x^2}}$ (13)
- (b) Calculate $\int e^x \left(\frac{1-x}{1+x^2}\right)^2 dx$ (10)
- (c) Calculate $\int \frac{dx}{a \sin x + b \cos x}$ (12)
- Q 2. (a) Obtain a reduction formula for $\int \sec^n x dx$ (11)
- (b) Evaluate $\int_0^1 \frac{\log(1+x)}{1+x^2} dx$ (12)
- (c) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{5+4 \sin x}$ (12)
- Q 3. (a) Evaluate $\lim_{n \rightarrow \infty} \left[\frac{1^2}{n^3+1^3} + \frac{2^2}{n^3+2^3} + \frac{3^2}{n^3+3^3} + \dots + \frac{n^2}{n^3+n^3} \right]$ (11)
- (b) Determine $\int_0^{\frac{\pi}{2}} \sin^n x dx$; when n is even and odd positive integers respectively. (12)
- (c) Define Beta function and Gamma function. Prove that, $\sqrt{\frac{1}{2}} = \sqrt{\pi}$ (12)
- Q 4. (a) Use a double integral to find the area enclosed between the curves, (11)
 $y^2 = 4ax$ and $x^2 = 4ay$
- (b) Evaluate $\iint_R \frac{x-y}{x+y} dA$ by making the change of variables, where R is the (13)
region enclosed by $x - y = 0$, $x - y = 1$, $x + y = 1$ and $x + y = 3$
- (c) Use a triple integral to find the volume of the solid enclosed between the (11)
cylinder $x^2 + y^2 = 9$ and the planes $z = 1$ and $x + z = 5$

Section B

(Answer ANY THREE questions from this section in script B)

- Q 5. (a) A function $f(x)$ is defined as follows: (13)

$$f(x) = \begin{cases} x, & \text{for } 0 < x < 1 \\ 2 - x, & \text{for } 1 \leq x \leq 2 \\ x - \frac{x^2}{2}, & \text{for } x > 2 \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x = 1$

- (b) Differentiate, $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ with respect to $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ (10)

- (c) Find the following value of limits: (12)

(i) $\lim_{x \rightarrow 0} \frac{\tan x - x}{x - \sin x}$

(ii) $\lim_{x \rightarrow 0} \frac{x \cos x - \log(1+2x)}{x^2}$

- Q 6. (a) Find the n 'th derivatives of $y = \frac{1}{6x^2 - 5x + 1}$ (12)

- (b) State Leibnitz's theorem. If $y = \sin^{-1} x$, then show that (13)

$$(1 + x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$$

- (c) State Rolle's theorem. Is Rolle's theorem applicable to $f(x) = \sin x$ in $(0, \pi)$? (10)

- Q 7. (a) Define homogeneous function and if, $V = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, then show that $xV_x + yV_y = \tan V$ (11)

- (b) If, $u = \log r$ and $r^2 = x^2 + y^2 + z^2$, then find the value of (12)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$

- (c) Find the angles of intersection of the curves, (12)

$$x^2 = 4y \text{ and } y(x^2 + 4) = 8$$

- Q 8. (a) Discuss the maxima and minima of (12)

$$f(x, y) = x^3 + y^3 - 3x - 12y + 20$$

- (b) Find the tangent and normal at (a, b) to the curve $\left(\frac{x}{a}\right)^3 + \left(\frac{y}{b}\right)^3 = 2$ (13)

- (c) Find the centre of curvature of the curve, $xy = x^2 + 4$ at $(2, 4)$ (10)

Khulna University of Engineering & Technology
 B.Sc. Engineering 1st year 1st term (Regular) Examination, 2017
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 Ch 1127
 Inorganic and Physical Chemistry

Time: 3 hours

Full Marks: 210

N.B.: (i) Answer **ANY THREE** questions from each section in separate script.

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Section A

(Answer **ANY THREE** questions from this section in **script A**)

- Q1. (a) What is bond order? Calculate the bond order in the following molecules: (10)



- (b) Illustrate the Lewis structure of i) H_2 , ii) N_2 , iii) SO_3 , iv) PO_4^{3-} , v) ClO_4^- , vi) NO_2^- and vii) O_2 (07)

- (c) Given the following information for Mg, O_2 , and MgO, calculate the second electron gain enthalpy for O_2 . (08)

For $Mg(s)$, $\Delta H_{sub} = +148 \text{ KJ/mol}$

1st ionization energy for $(Mg) = +738 \text{ KJ/mol}$

2nd ionization energy for $(Mg) = +1450 \text{ KJ/mol}$

Bond dissociation for $O_2 = +499 \text{ KJ/mol}$

1st electron gain for $O_2 = -141 \text{ KJ/mol}$

For MgO , Lattice energy = $+3890 \text{ KJ/mol}$

For MgO , Enthalpy of formation = -602 KJ/mol

- (d) Distinguish between intermolecular and intramolecular hydrogen bonding with examples? Write down the applications of H-bonding. (10)

- Q2. (a) Define "diffusion layer" in electrochemistry. "The feasibility of a reaction can be predicted from the measurement of emf of a cell" – explain. (11)

- (b) For the following cell (07)



Given, $E_{Fe^{2+}, Fe}^0 = -0.44 \text{ V at } 298^\circ K$

$E_{Cr^{3+}, Cr} = -0.71 \text{ V at } 298^\circ K$

Calculate Δh for the cell reaction at $298^\circ K$

- (c) Write short notes on i) Transport number and (ii) Polarization (10)

- (d) Sketch a complete diagram of a 3 electrodes electrochemical cell. (07)

- Q3. (a) Define molecularity and order of a reaction. Why the reactions of higher order are rare? (10)

- (b) What does mean 'steady state approximation'? Why this approximation is superior to collision theory, explain? (10)

- (c) If a unimolecular reaction is one quarter completed in 100 minutes, what is the specific rate constant, k and period of half-life, where reaction rate depends on single molecule? (08)

- (d) Explain with examples, the function of catalytic promoters and poisons in chemical reaction. (07)

- Q4. (a) What is electrochemical corrosion? Explain the various factors influencing the rate of corrosion (10)
- (b) "Corrosion can be protected cathodically"- Explain. (08)
- (c) Write short note on Ionic crystal. (05)
- (d) What are primary and secondary reference electrode? Mention an electrode reaction where the saturated calomel electrode acts as cathode? Write down the drawbacks of standard hydrogen electrode. (12)

Section B

(Answer **ANY THREE** questions from this section in **script B**)

- Q5. (a) Define radial distribution function? Discuss the radial distribution curve for water at 1.5 °C. (12)
- (b) Define and explain the term "Solution"? When two components A and B are mixed and formed an ideal solution, determine the entropy change of mixing thermodynamically. (12)
- (c) State and derive Raoult's law thermodynamically. (11)
- Q6. (a) Explain the terms (i) Degree of freedom, (ii) Phase, and (iii) Component. (12)
- (b) Discuss the phase diagram of water. Explain the triple point is an invariant point. (15)
- (c) Define CST? Water and nicotine system has UCST at 210 °C and LCST at 60 °C, draw the phase diagram of the mixture. (08)
- Q7. (a) What are colloids? How are they classified? (06)
- (b) Discuss the electrodialysis method for purification of colloids. (08)
- (c) Define zeta potential? Outline the principles of Stern's theory for the structure of the electric double layer. (15)
- (d) Distinguish between electrophoresis and electroosmosis? (06)
- Q8. (a) Define 'Solid solution'. Explain why 'normality' and 'molarity' change with temperature but 'molality' and 'mole fraction' are independent of temperature (12)
- (b) 'Mixture of alcohol and water can not be separated by fractional distillation'- explain (08)
- (c) Explain the following reason: (08)
- (i) Solubility of a gas in a given liquid decreases with rise in temperature.
- (ii) A salt solution fails to quench thirst.
- (d) Calculate the ionic strength of a solution containing 0.2 M NaCl and 0.1 M Na₂SO₄ (07)

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EEE 1127

Electrical Engineering Fundamentals

Time: 3 hours

Full Marks: 210

- N.B: (i) Answer **ANY THREE** questions from each section in separate scripts.
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Section A

(Answer **ANY THREE** questions from this section in **script A**)

- Q 1. (a) State and explain Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL). (12)
Calculate the value of V_{ab} , V_b and V_c of the following Fig. Q 1(a).

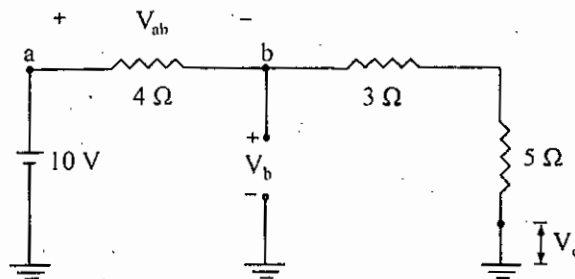


Fig. Q 1(a)

- (b) Define Branch, Node and Mesh. Find the current through each branch of the network (12)
showing below (Fig. Q 1(b)) using Mesh analysis.

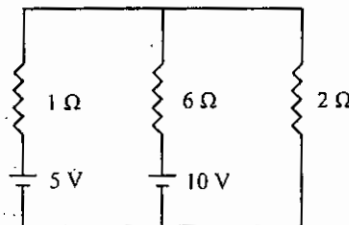


Fig. Q 1(b)

- (c) State superposition theorem. Determine the current through 4Ω resistor using this (11)
theorem for the network shown in Fig. Q 1(c)

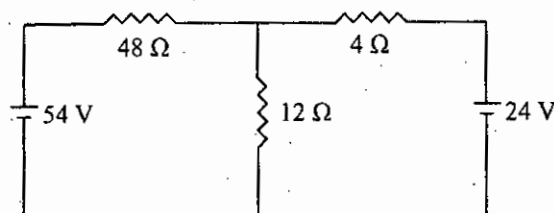


Fig. Q 1(c)

- Q 2. (a) With proper diagram deduce the condition for maximum power transfer. Also write (12)
down the equations for maximum power and efficiency during this condition.
- (b) Define (i) Period (ii) Cycle (iii) Angular velocity (iv) RMS value (v) Average value. (12)
Find the value of amplitude, phase, period and frequency of the sinusoid,
 $V(t) = 12 \cos(40t + 20^\circ)$
- (c) Show that the crest factor of a sine wave is $\sqrt{2}$ and the form factor is 1.11. (11)
- Q 3. (a) State and explain Ampere's circuital law. Define magnetic field intensity and (10)
magnetic flux density.
- (b) Define permeability, magnetomotive force and reluctance. Compare between electric (10)
circuit and magnetic circuit.
- (c) Define measuring instrument and classify them. How can you extend the range of (08)
ammeter and voltmeter.
- (d) A moving coil instrument gives a full scale deflection of 10 mA, when the potential (07)
difference across its terminal is 100 mV. Calculate
- (i) The shunt resistance for a full scale deflection corresponding to 100 mA.
- (ii) Resistance for full scale reading with 1000 V

- Q 4. (a) Define impedance. Find the impedance and power absorbed by a RL Branch and also show them graphically. (13)
- (b) What is the condition of series resonance in a circuit? Explain with necessary phasor diagram. (08)
- (c) Define power factor. For the circuit shown in Fig. Q 3(c), calculate real power, reactive power, apparent power, power factor and also draw the complete vector diagram of this circuit. (14)

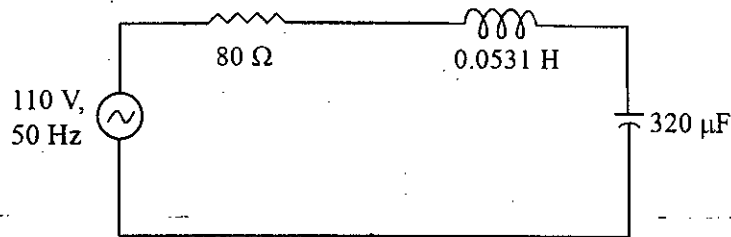


Fig. Q 4(c)

Section B

(Answer ANY THREE questions from this section in script B)

- Q 5. (a) Derive the E.M.F equation of a DC generator. (10)
- (b) Write down the functions of yoke and commutator. Draw the open circuit characteristic (O.C.C) and external characteristic of a DC shunt generator. (08)
- (c) Define critical resistance. Explain the voltage build-up process for self-excited DC generator. (10)
- (d) Define armature reaction. The coil of a four pole generator is rotating at a speed of 1200 RPM. The flux per pole is 2.5×10^{-5} Wb. Find the voltage induced in the coil, if it has 100 turns. (07)
- Q 6. (a) Define counter E.M.F. Derive the speed equations for both DC shunt and DC series motor. (12)
- (b) Draw the speed-torque characteristic and electrical characteristic of DC series motor. Explain from the characteristics, why DC series motors are never used unless they are directly connected to the load. (10)
- (c) What is the necessity of a starter? Describe a 3-point starter. Write down the relevant features of a 4-point starter over 3-point starter. (13)
- Q 7. (a) Describe the working principle of a single phase transformer and also draw the exact equivalent circuit. (12)
- (b) The 25 KVA transformer has 500 turns on the primary and 50 turns on the secondary. The primary is connected to 3000 V, 50 Hz supply. Find the full-load primary and secondary currents, the secondary E.M.F and the maximum flux in the core. Neglect leakage drops and no-load primary current. (12)
- (c) Derive the condition for maximum efficiency of a transformer. Why transformer rating is KVA? (11)
- Q 8. (a) How rotating field is developed in Induction motor for three phase supply? Describe with neat sketch. (12)
- (b) Draw the complete torque-speed curve of an induction motor. (05)
- (c) Define slip. A 3Φ induction motor is wound for 4 poles and is supplied from 50 Hz system. Calculate (08)
- Synchronous speed
 - Rotor speed when slip is 4% and
 - Rotor circuit frequency when rotor runs at 1000 RPM.
- (d) Describe the basic working principle of an alternator. (10)